

Species-Aware Review: Folate One-Carbon Carrier-State Interconversion in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module area:** cofactors_vitamins_redox **Local bucket:** kegg:ppu00670 "One carbon pool by folate" (31 candidate genes) **Generic module:** Folate one-carbon carrier-state interconversion (4 chemically distinct reactions)

1. Executive Summary

The folate carrier-state interconversion module is **fully satisfiable (covered)** in *P. putida* KT2440. All four reactions of the generic module have high-confidence encoding:

- **Required core** (methylene-THF $\rightleftharpoons$ methenyl-THF redox + methenyl-THF $\rightarrow$ 10-formyl-THF hydrolysis): **redundantly covered** by two genuinely bifunctional Fold paralogs, **fold1/PP_1945** and **fold2/PP_2265** (both carry the complete NADP-linked dehydrogenase EC 1.5.1.5 + cyclohydrolase EC 3.5.4.9 domain architecture).
- **Optional MTHFR branch** (methylene-THF $\rightarrow$ 5-methyl-THF): **covered** by compact NADH-type **metF/PP_4977** (EC 1.5.1.54), with a second, architecturally distinct MTHFR-family paralog **PP_4638** (candidate_uncertain).
- **Optional 5-formyl-THF salvage branch:** **covered** by **fau/PP_5203** (EC 6.3.3.2).

The dominant curation issue is **not a gap but over-inclusion**: the KEGG ppu00670 bucket returns exactly the 31 candidates, but only 4–5 of them are true carrier-state-interconversion genes. The other ~24 are upstream one-carbon **loading** enzymes (glyA SHMT, glycine-cleavage system), **downstream use** enzymes (purN, purH, thyA, metH, metK, ahcY), folate-pool **regeneration** (folA, folM), a 10-formyl-THF **drain** (purU $\times$ 3), and **osmolyte/sulfur** enzymes (betA, betB, PP_0708, PP_4594) that are not folate carrier-state reactions at all. KT2440 uses the fused Fold architecture exclusively; no MtdA/FchA split or eukaryotic MTHFD is expected. Evidence is homology-based (PE3) for the core genes, with protein-level evidence (PE1) only for two of the peripheral PurU deformylases.

2. Target-Organism Pathway Definition

Included process (this module): the interconversion of substituted-THF carrier states — 5,10-methylene-THF, 5,10-methenyl-THF, 10-formyl-THF, 5-methyl-THF, and the salvaged 5-formyl-THF pool — via four chemically distinct transformations:

1. NAD(P)-linked oxidation of 5,10-methylene-THF → 5,10-methenyl-THF (dehydrogenase).
2. Hydrolysis of 5,10-methenyl-THF → 10-formyl-THF (cyclohydrolase). (*Coupled with #1 as the required core.*)
3. Reduction of 5,10-methylene-THF → 5-methyl-THF (MTHFR). (*Optional branch.*)
4. ATP-dependent salvage of 5-formyl-THF → 5,10-methenyl-THF (cyclo-ligase). (*Optional branch.*)

Neighboring pathways to keep separate (present in KEGG ppu00670 but **outside** this module): - **One-carbon loading:** serine hydroxymethyltransferase (glyA1/glyA2), glycine cleavage system (gcvT/P/H-I&II, lpd/lpdG/lpdV), and formate-THF ligase (Fhs — not encoded in KT2440). → KEGG ppu00260/ppu00630/ppu00785. - **One-carbon use:** purine synthesis (purN, purH), thymidylate synthase (thyA), methionine synthase (metH) and the SAM/methionine cycle (metK, ahcY). → KEGG ppu00230/ppu00240/ppu00270/ppu00450/ppu00920. - **Folate/pterin pool regeneration:** DHFR (folA), dihydromonapterin reductase (folM). → folate biosynthesis ppu00790. - **Osmolyte and sulfur metabolism:** choline→glycine-betaine oxidation (betA, betB, PP_0708) and cystathionine γ -synthase (PP_4594). → glycine-betaine biosynthesis / cysteine-methionine metabolism; erroneously mapped here.

Alternate names / database definitions: KEGG map 00670 "One carbon pool by folate"; the module core overlaps MetaCyc "folate transformations" / "5,10-methylenetetrahydrofolate interconversion"; the fused enzyme is annotated FOLD (bacterial) or D/C domain of MTHFD1 (eukaryotic). Note the EC-number mismatch in the local metadata for FOLD: the dehydrogenase should read EC **1.5.1.5** (NADP+) and cyclohydrolase EC **3.5.4.9**, as confirmed in UniProt.

3. Expected Step Model (with target-taxon status)

Module step	Reaction	Expected enzyme family	KT2440 status	Gene(s)
Core dehydrogenase	5,10-CH ₂ -THF → 5,10-CH ⁺ -THF (NADP ⁺)	Bifunctional FolD	covered	fold1/ PP_1945, fold2/ PP_2265
Core cyclohydrolase	5,10-CH ⁺ -THF → 10-CHO-THF	Bifunctional FolD (fused)	covered	fold1/ PP_1945, fold2/ PP_2265
MTHFR branch (optional)	5,10-CH ₂ -THF → 5-CH ₃ -THF	Compact NADH MetF	covered	metF/ PP_4977
MTHFR branch alt. architecture	(regulatory-architecture MTHFR)	MTHFR + extra domain	candidate_uncertain	PP_4638
5-formyl-THF salvage (optional)	5-CHO-THF → 5,10-CH ⁺ -THF (ATP)	Fau/MTHFS cyclo-ligase	covered	fau/ PP_5203
MtdA monofunctional dehydrogenase	(methylotroph alternative)	MtdA	not_expected_in_target_taxon	—
FchA monofunctional cyclohydrolase	(Clostridial alternative)	FchA	not_expected_in_target_taxon	—
Eukaryotic MTHFD1/ MTHFD2	(eukaryotic/ mitochondrial)	multidomain	not_expected_in_target_taxon	—

4. Candidate Genes and Evidence

4A. True module members (high confidence)

- **fold1 / PP_1945 (Q88LI7, 291 aa, PE3, score 3).** Bifunctional Fold. InterPro IPR000672 THF_DH/CycHdrlase; Pfam PF00763 + PF02882 (both catalytic and NAD(P)-binding domains). UniProt "includes" both EC 1.5.1.5 and EC 3.5.4.9. Covers core dehydrogenase **and** cyclohydrolase. *Caveat*: homology-based; not experimentally characterized in KT2440.
- **fold2 / PP_2265 (Q88KM5, 284 aa, PE3, score 3).** Second bifunctional Fold with identical domain content. Provides redundancy. *Curation note*: two full-length Fold paralogs — treat both as covered; a later demonstration that one is dominant does not create a gap.
- **metF / PP_4977 (Q88D51, 295 aa, PE3, EC 1.5.1.54).** Compact bacterial MTHFR — InterPro IPR004620 (bacterial-MTHFR signature), Pfam PF02219, FAD flavoprotein, NAD-linked. Matches the "NADH-linked compact MetF" module variant. Genomically clustered with metH-cycle genes (metK PP_4967, ahcY PP_4976) consistent with a methionine-synthesis role.
- **fau / PP_5203 (Q88CH8, 202 aa, PE3, EC 6.3.3.2).** 5-formyltetrahydrofolate cyclo-ligase; Pfam PF01812. Covers the optional salvage branch (recycles the "folate trap" metabolite 5-CHO-THF back to 5,10-CH⁺-THF). *Caveat*: the local metadata lists its primary bucket as ppu04981 (folate biosynthesis map), a minor cross-mapping discrepancy; function is unambiguous.

4B. Ambiguous / promote to review

- **PP_4638 (Q88E30, 495 aa, PE3, no EC, no gene symbol).** Genuine MTHFR-catalytic domain (Pfam PF02219 / IPR003171; SUPFAM SSF51730; FAD flavoprotein) **fused to a C-terminal DUF5981** (Pfam PF12225 / IPR022026). It **lacks** the bacterial-MTHFR signature IPR004620 that metF carries. This is the module's "regulatory-architecture MTHFR" alternative implementation — a real second MTHFR-family protein, but its true substrate/cofactor and whether it reduces 5,10-methylene-THF are unconfirmed. **Do not** silently merge with metF or dismiss as noise.

4C. Adjacent (present in KEGG map, outside module)

- **purU-I/PP_0327 (Q88R07, PE1), purU-II/PP_1367 (Q88N49, PE3), purU-III/PP_1943 (Q88LI9, PE1)** — formyltetrahydrofolate deformylase, EC 3.5.1.10 (10-CHO-THF → formate + THF). A **drain** of the formyl pool, not a carrier-state interconversion. Triplication with two

PE1 (protein-level) copies signals physiologically important formate/one-carbon overflow in *P. putida*. *purU*-III is genomically adjacent to *fold1*. → adjacent module, not this one.

- **glyA1/PP_0322, glyA2/PP_0671** (SHMT, EC 2.1.2.1) — one-carbon **loading**. Excluded.
- **gcvT/P/H-I & II (PP_0986/0988/0989, PP_5192/5193/5194), lpd/lpdG/lpdV (PP_5366/4187/4404)** — glycine cleavage system + shared E3 dihydrolipoyl dehydrogenase — one-carbon **loading**. Excluded (*lpd* genes are broadly shared with 2-oxo-acid dehydrogenases; classic over-propagation into folate maps).
- **purN/PP_1664 (EC 2.1.2.2), purH/PP_4822 (EC 2.1.2.3/3.5.4.10)** — 10-CHO-THF-dependent purine synthesis; **downstream use**. Excluded.
- **thyA/PP_5141 (EC 2.1.1.45)** — thymidylate synthase, consumes 5,10-CH₂-THF; **downstream use**. Excluded.
- **metH/PP_2375 (EC 2.1.1.13), metK/PP_4967 (EC 2.5.1.6), ahcY/PP_4976 (EC 3.13.2.1)** — cobalamin-dependent methionine synthase + SAM/SAH cycle; **downstream use** of 5-CH₃-THF. Excluded.
- **folA/PP_5132 (EC 1.5.1.3), folM/PP_4632 (EC 1.5.1.3/1.5.1.50)** — DHFR / dihydromonapterin reductase; **folate/pterin pool regeneration**. Excluded.
- **betA/PP_5064, betB/PP_5063, PP_0708 (betaine-aldehyde DH), PP_4594 (cystathionine γ-synthase)** — glycine-betaine osmolyte biosynthesis and sulfur-amino-acid metabolism; **not folate reactions**. Clear over-inclusion by the KEGG bucket.

5. Gaps, Ambiguities, and Likely Over-Annotations

- **No true gaps.** Every required and optional module reaction is encoded.
- **Over-annotation into the bucket (primary issue):** ~24 of 31 candidates are loading/use/regeneration/osmolyte genes. *betA/betB/PP_0708/PP_4594* are the most clear-cut misassignments to a folate carrier-state module. The *lpd/lpdG/lpdV* E3 subunits are generically shared and should not anchor any folate-specific claim.
- **Paralog ambiguity:** two *Fold* copies and three *PurU* copies. *Fold* redundancy is benign for satisfiability; *PurU* triplication is a boundary question (drain vs core).
- **PP_4638:** distinct-architecture MTHFR-family protein without EC assignment — the single genuine "candidate_uncertain" for the in-module steps.

- **Metadata EC discrepancy:** Fold dehydrogenase EC should be 1.5.1.5 (NADP+), matching UniProt; some local strings blur EC granularity. fau's primary-bucket cross-mapping (ppu04981 vs ppu00670) is a minor inconsistency.
- **Evidence depth:** core-module conclusions rest on **homology (PE3)**, not KT2440 biochemistry. Only PurU-I and PurU-III carry protein-level evidence, and that is presence, not mechanism.
- **PP_4638 is uncharacterizable by transfer:** of 1,230 UniProt proteins carrying the DUF5981 domain (PF12225), **zero are reviewed/experimentally characterized**, so no functionally validated homolog exists for the MTHFR+DUF5981 architecture anywhere. Its EC-less annotation therefore cannot be resolved by homology alone.
- **No alternative formyl-pool entry:** proteome UP000000556 encodes **no formate-tetrahydrofolate ligase Fhs (EC 6.3.4.3)**. The only routes into the 10-formyl-THF/methenyl-THF pool are the Fold core and fau salvage; purU drains it to formate. This makes the Fold core effectively non-redundant at the pathway level (redundant only in gene copy number).
- **MTHFR branch is B12-coupled:** KT2440 encodes **only cobalamin-dependent MetH (EC 2.1.1.13)** and **no cobalamin-independent MetE (EC 2.1.1.14)**, so the optional 5-methyl-THF branch product is consumed exclusively via a B12-dependent step (context for whether the metF/PP_4638 branch is conditionally essential).

6. Module and GO-Curation Recommendations

Per-step module status: - Core dehydrogenase ($5,10\text{-CH}_2\text{-THF} \rightarrow \text{CH}^+\text{-THF}$): **covered** (foldD1, foldD2). - Core cyclohydrolase ($\text{CH}^+\text{-THF} \rightarrow 10\text{-CHO-THF}$): **covered** (foldD1, foldD2). - MTHFR branch: **covered** (metF); PP_4638 **candidate_uncertain**. - 5-formyl-THF salvage branch: **covered** (fau). - MtdA / FchA / eukaryotic MTHFD variants: **not_expected_in_target_taxon**.

Boundary / module-document recommendations: - Flag the local bucket as **module_needs_revision at the bucket level**: kegg:ppu00670 conflates loading + interconversion + use + salvage + osmolyte metabolism. For this module only foldD1/foldD2, metF (+PP_4638), and fau should be scored as members. - Consider a **separate "10-formyl-THF drain / formate generation" annotation** for purU-I/II/III rather than counting them in carrier-state interconversion. - Record **Fhs (formate-THF ligase)** as **not_expected_in_target_taxon** for the loading side; note there is no alternative ATP-dependent formate entry into the formyl pool in KT2440. - Recommend GO assignments: foldD1/foldD2 $\rightarrow$ GO:0004488 (methylenetetrahydrofolate dehydrogenase (NADP+)) **and** GO:0004477

(methenyltetrahydrofolate cyclohydrolase); metF → GO:0004489 (methylenetetrahydrofolate reductase [NAD(P)H]); fau → GO:0030272 (5-formyltetrahydrofolate cyclo-ligase). **Do not** propagate a methylenetetrahydrofolate-reductase GO to PP_4638 without functional review; a "MTHFR-domain, function-uncertain" hold is appropriate. No new GO term request is obviously required.

7. Genes to Promote to Full fetch-gene Review

1. **PP_4638 (Q88E30)** — highest priority. Resolve whether the MTHFR+DUF5981 fusion is a bona fide second methylenetetrahydrofolate reductase (cofactor? activity?) or a divergent flavoenzyme; determines whether the MTHFR step has one or two implementations.
2. **fold1/PP_1945 vs fold2/PP_2265** — confirm both are functional (not one pseudogene/paralog-subfunctionalized); check expression/essentiality context and which is the housekeeping copy.
3. **metF/PP_4977** — confirm NADH vs NADPH preference and operon context to lock the "compact MetF (NADH)" variant assignment.
4. **purU-I/II/III (PP_0327/PP_1367/PP_1943)** — clarify boundary role (formate overflow) and whether they warrant a separate module; leverage the existing PE1 evidence.

8. Key References

- Belda E, *et al.* (2016) The revisited genome of *Pseudomonas putida* KT2440 enlightens its value as a robust metabolic chassis. *Environ Microbiol* 18:3403–3424.

P 26913973

(KT2440 re-annotation; basis for gene calls.)

- Aluri S, *et al.* (2016) Physiological role of Fold, FchA and Fhs from *Clostridium perfringens* in a heterologous *E. coli* model. *FEBS J*

P 26531681

(Fused Fold vs split monofunctional Fold-dehydrogenase + FchA-cyclohydrolase architectures — supports "not_expected" call for FchA in *P. putida*.)

- Eadsforth TC, *et al.* (2012) *Acinetobacter baumannii* FolD ligand complexes.

P 23050773

(Bifunctional dehydrogenase/cyclohydrolase FolD is widely distributed in Gram-negative bacteria — transfer to *P. putida* strong.)

- Goenrich M, *et al.* (2002) Methylene-THF dehydrogenase FolD from *Hyphomicrobium* / MtdA in methylotrophs.

P 11889483

(Distinguishes classical bifunctional FolD from methylotroph-specific MtdA.)

- Pawelek PD & MacKenzie RE (1998) Cyclohydrolase is rate-limiting in bifunctional D/C enzymes.

P 9454603

(Mechanistic coupling of the two core activities.)

- Sah S, *et al.* (2020) Monomeric NADH-oxidizing MetF/MTHFR from *Mycobacterium smegmatis*.

P 32253341

(Bacterial MetF diversity, relevant to metF variant assignment.)

Evidence transfer note: All core in-target conclusions are **inferred from homology/domain architecture (UniProt/InterPro PE3)** plus the KT2440 re-annotation genome; they are **not** backed by direct KT2440 enzymology. Cross-organism enzyme references (Gram-negative FolD, bacterial MetF) transfer **strongly** to *P. putida* on sequence/domain grounds. PP_4638's activity is the one substantive open question.